Unsupervised Detection of Topological Thermal Transitions via Symmetry-Invariant Diffusion Maps

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Draft Report
(Dated: December 21, 2025)

Detecting and classifying phases of matter from sampled configurations is challenging when an order parameter is unknown or absent, as in topological transitions. A minimal, fully unsupervised pipeline based on diffusion maps and simple clustering is evaluated as a baseline for locating thermal phase transitions. Special distance metrics are constructed to respect the system's symmetries. The analysis focuses on the 2D Ising model (with comparison to the 2D XY setting) and emphasizes interpretability: a low-dimensional embedding is produced, along with scalar diagnostics derived from clustering and the spectral properties of the associated diffusion operator. Finite-size behavior is assessed by repeating the pipeline across multiple lattice sizes.

I. INTRODUCTION

Mapping out the phase space of a desired material is a very active area of research in both statistical physics and condensed matter physics. In recent years, machine learning techniques have been employed to classify and probe more complicated models. In classical models, data can often be largely synthetically generated using Monte Carlo techniques. Supervised machine learning techniques use this data, along with labels corresponding to the parameter space, to correctly classify the phase. Recently, unsupervised techniques have also been explored. These methods do not need prior knowledge of the system and can provide unbiased classification. They typically work to convert a high-dimensional input lattice configuration into a lower-dimensional, interpretable representation [2, 11]. However, much of the modern machine-learning literature on phase transitions is difficult to evaluate rigorously in practice. Representative examples include supervised classifiers and hybrid schemes that still rely on labeled training tasks to locate a critical temperature. Purely unsupervised baselines can succeed on the Ising model by isolating a component related to an observable (such as magnetization or energy), but do not necessarily transfer to settings where the relevant structure is more subtle, as in the 2D XY model. Common weaknesses that appear repeatedly (often implicitly) include:

- **Hidden supervision:** Hyperparameters are tuned until the output matches a known answer.
- **Limited scaling checks:** Results are reported for a small subset of lattice sizes, risking strong finite-size effects [1].
- **Low interpretability:** Large neural networks can achieve high supervised accuracy but require labeled data and can be difficult to interpret [2, 11].

Diffusion maps have recently become popular as an unsupervised classification scheme. Input data is transformed via a process called diffusion to produce a low-dimensional representation. Recent diffusion-based methods have achieved great success in identifying phase transition boundaries even in multi-phase systems. However, the current literature often limits the use of diffusion maps to detect topological differences between ground states. Additionally, the use of a Euclidean metric is the de facto choice, with only modifications done to the hyperparameter ε. This, however, presents a problem for many models whose underlying physics remains invariant under global transformations, resulting in redundant information in the diffusion step. Recent work [15] has presented diffusion kernels that are so-called G-invariant. Using this, we shall present a new minimal diffusion-based kernel approach that illuminates the transition for models that remain difficult to analyze with other techniques, such as the XY model. Accordingly, the analysis focuses on a minimal and interpretable baseline that (i) is unsupervised at fit time, (ii) produces an explicit low-dimensional embedding, and (iii) can be tested across lattice sizes. The outline is as follows. Section II reviews related work and limitations. Sections III–IV provide the necessary technical background, while Section V describes the methodology, including data generation and diffusion-map construction. Section VI presents results, and Sections VII–VIII discuss limitations and summarize conclusions.

II. RELATED WORK AND LIMITATIONS

Modern applications of machine learning to phase classification exhibit limitations that motivate an explicit emphasis on interoperability, minimal supervision, and finite-size checks.

a. Supervised Phase Classification: Neural networks can classify both Ising and XY configurations with high accuracy when trained on labeled data [3, 2]. This is useful, but it does not solve the main problem of phase discovery when labels or a known order parameter are unavailable. Given *unlabeled* data, a diagnostic is required that (a) points to a transition and (b) can be checked with scaling. A classifier can also succeed by including correlation bias (for example, by using a variable dependent on energy or magnetization that is specific to the dataset) [18].

b. Computational Demands: The confusion scheme [17] trains a classifier many times with different assumed label splits and looks for a characteristic performance curve. This requires repeatedly building and training a supervised model, choosing an architecture, and selecting hyperparameters. In practice, those choices can easily compound into a large computational workload and are not generally practical for more complicated models in higher-dimensional parameter spaces with more exotic phase boundaries.

c. Lack of Generalization: On the Ising model, even very simple unsupervised tools (PCA, k-means, etc.) can separate phases [18]. This is expected: the ordered phase differs from the disordered phase. In a global way (via magnetization), many algorithms will rediscover that structure. A key limitation is generalization: success on Ising does not imply success on transitions with topological, non-local, or strongly masked signals, or on those strongly affected by finite-size effects [11].

d. Symmetrical Configurations: Systems under close study also exhibit symmetries that leave the system's dynamics unchanged. This provides additional structural information for the ML model to learn, but can also pollute and obscure the phase information. Current literature often either doesn't directly address this or performs data preprocessing to handle it. However, such data methods don't fully address the underlying issue and can be difficult to implement for more complex site data.

This approach retains the core idea of geometry-based unsupervised learning: treat configurations as points in a high-dimensional space and infer structure from their geometric relationships. By constructing distance metrics tailored to the system, we can quotient out latent symmetries to ensure maximal dimensional reduction. To avoid over-interpreting a single visualization, the analysis includes explicit diagnostics and a finite-size check.

III. CLASSICAL MODELS

A. General Model

Classical models in statistical physics seek to describe a canonical ensemble in thermal equilibrium with a heat bath. Let $G = (V, E)$ be a graph with sites V and neighbor edges E, together with a single-site state space Ω. A configuration is then an assignment $\sigma \in \Omega^V$ assigning a state to each site in the graph. Such a model is governed by a complex collection of interactions $\Phi = (\Phi_A)_{A\subseteq V}$ where each interaction is represented by a function $\Phi_A : \Omega^A \to \mathbb{R}$. For a finite region $\Lambda \subset V$ we may introduce the total energy

$$H(\sigma_\Lambda) = \sum_{A\subset\Lambda} \Phi_A(\sigma_A) \tag{1}$$

For the more common models we study in this paper, we typically choose $V = \mathbb{Z}^d$ and $\Phi = \Phi_{\{i,j\}}$ to include only nearest-neighbor interactions. The Boltzmann distribution $P(\sigma_\Lambda) = \frac{1}{\mathcal{Z}} e^{-H/T}$ then gives the probability of a specific configuration occurring at temperature T, where $\mathcal{Z}$ is a constant chosen to ensure a probability measure.

B. 2D Ising Model

The 2D Ising model places a spin $s_i \in \{-1, +1\}$ on each site of an $L \times L$ square lattice. Neighboring spins prefer to align, with energy

$$H(s) = -J \sum_{\langle i,j\rangle} s_i s_j, \tag{2}$$

where $\langle i, j\rangle$ runs over nearest neighbors. This simple model has been studied for nearly a century [6]. At low temperature T, most spins point the same way (an *ordered* phase). At high T, thermal noise dominates and spins look random (a *disordered* phase). In the infinite system, the 2D Ising model has a sharp phase transition at $T_c = 2/\ln(1+\sqrt{2}) \approx 2.269$ [14]. In a finite lattice, the transition is smeared: there is no true singularity, only a rapid crossover. As L grows, the crossover becomes sharper and its midpoint shifts toward T_c [1]. Any method that claims to detect T_c should therefore be checked across multiple L values.

C. 2D XY Model

The 2D XY Model places an angular spin $\theta_i \in S^1$ on each site of an $L \times L$ square lattice. Neighboring spins similarly prefer to align, with energy

$$H(\theta) = -J \sum_{\langle i,j\rangle} \cos(\theta_i - \theta_j) \tag{3}$$

This model can be extended to any-dimensional hypercube, although a special kind of phase transition occurs only in the 2-dimensional case. Equilibrium configurations occur when either all the spins are aligned or with vortices/anti-vortices. At low T, thermal excitations become too weak, and only vortex-antivortex pairs can occur. At sufficiently high T, however, vortices may become unbidden and proliferate throughout the lattice. This transition is known as a Berezinskii-Kosterlitz-Thouless (BKT) transition, and numerical simulations for the infinite lattice put the critical temperature at $T_c \approx 0.8935$. This model suffers from severe finite-size effects, making it very computationally expensive to probe and study.

IV. DIFFUSION MAPS

Diffusion maps are a geometric-based way to compress high-dimensional data to a few meaningful coordinates [4, 5]. The idea stems from manifold learning, where the assumed data collection lies in a submanifold of the high-dimensional space.

A. Lattice Diffusion

Here the input is a set of configurations $x_1, \ldots, x_N$ where $x_i \in \mathbb{R}^m$ with $m = L \times L$. Given a distance metric $d : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}$, the kernel matrix K is defined as

$$K_{ij} = \exp\big(-d(x_i, x_j)^2/\varepsilon\big), \tag{4}$$

where ε controls what counts as "similar." If ε is too small, the manifold breaks into tiny pieces. If ε is too large, everything looks similar and the method cannot separate structure. After a standard normalization [4] described by relation

$$P_{ij} = \frac{K_{ij}}{\sum_j K_{ij}} \tag{5}$$

the matrix becomes a one-step transition rule for a random walk on the data. Intuitively, the walk moves easily inside a dense region (many similar configurations), but it is slow to move between two regions separated by a low-density "bottleneck." The eigenvalues satisfy $1 = \lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \cdots$. The first eigenvector is constant (it carries no information). The next eigenvectors provide coordinates

$$\Psi_k(x_i) = \lambda_k \, \psi_k(x_i), \tag{6}$$

of which a subset $k \leq n$ are used. Large gaps like $\lambda_{k-1} - \lambda_k$ are a common sign of a clean cluster structure (related to spectral clustering) [13, 9].

B. Symmetry Invariant Diffusion

In practice, the choice of a distance metric $d(\cdot, \cdot)$ is important in conjunction with a proper choice of ϵ. This is because together they decide how the diffusion step spreads across the manifold. In fact, one can even use a so-called **pseudometric**. This is a metric where points can have a null distance separation even though they remain different. This is useful if we want to study spaces that locally resemble a quotient space. Such spaces appear under a finite group action and can locally resemble $\mathbb{R}^m \backslash G$, also known as an orbifold. The resulting quotient space is of lower dimension than the entire space $\mathbb{R}^m$, and can filter out any redundant symmetry information. Recent works [15] have attempted to fully derive and study special diffusion map embeddings that are invariant under the action of general compact Lie groups. Such approaches can numerically generate distances that resemble those of a pseudometric and can reduce the resulting embedding dimension. Proofs have also been provided showing that the resulting graph Laplacian converges to the Laplace-Beltrami operator Δ_G in the limit. In practice, such extensive machinery is not needed, and our approach constructs minimal pseudometrics that remain invariant under the group actions.

V. METHODS

A. Data generation

Equilibrium samples are generated on an $L \times L$ lattice with periodic boundary conditions using Markov chain Monte Carlo (MCMC). Single-spin Metropolis updates are used away from criticality [12], while Wolff cluster updates are applied in a narrow temperature window around T_c to mitigate critical slowing down [19]. A *sweep* is defined as L^2 attempted single-spin updates.

For each temperature on a dense grid spanning both phases, a fixed thermalization period is followed by recorded configurations separated by a fixed number of sweeps (thinning) to reduce temporal correlations. Basic per-sample observables (absolute magnetization and energy) are retained for sanity checks.

B. Unsupervised analysis

The unsupervised analysis proceeds as follows:

1. Represent each configuration as a vector $x_i \in \mathbb{R}^m$ and optionally *block-average* spins to reduce noise and dimension.
2. Choose a symmetry-aware squared distance $d^2(x_i, x_j)$.
3. Determine the kernel bandwidth via the heuristic: $\varepsilon = \mathrm{median}\{d^2(x_i, x_j)\}_{i<j}$.
4. Build a Gaussian kernel $K_{ij} = \exp(-d^2(x_i, x_j)/\varepsilon)$ and row-normalize it to a Markov matrix P.
5. Compute diffusion coordinates from the leading nontrivial eigenvectors of the associated symmetric normalization (Section IV). This is determined by finding the largest gap $\lambda_k - \lambda_{k+1}$ between eigenvalues, and only taking the corresponding k diffusion coordinates.
6. Estimate the critical temperature from the collection of diffusion coordinates, without using labels.

The final result clusters in the embedding using k-means [8, 10]. There are two important diagnostic metrics we may use to analyze the corresponding diffusion space.

- By looking at the first k diffusion coordinates, we can build a surface $\Psi(x) = [\Psi_1(x), ..., \Psi_k(x)]$ within embedding space $\mathbb{R}^k$ representing lattice configurations. This surface will have hills and valleys, which correspond to the accumulation of points on a specific phase. Regions where the diffusion coordinates change most rapidly correspond to phase transitions. Therefore, by analyzing $||\nabla\Psi||$, we may find possible phase transitions.

- Using the k-means clustering results, we can determine, for each temperature, the proportion of samples in each cluster. The critical temperature occurs when there is maximal ambiguity in between clusters, ie, when half of the samples are in each cluster. Therefore, by plotting the ordered fraction against the temperature, we can determine when it reaches $\frac{1}{2}$ and identify the phase transition point.

The silhouette score [16] and spectral quantities like the eigengap $\lambda_k - \lambda_{k+1}$ and mixing proxy $1 - \lambda_1$ are used to check whether the embedding has a clear distinguishable structure.

C. Kernel-scale selection

The diffusion kernel bandwidth ε determines the locality of the similarity graph. Typically, ε is chosen to be the median heuristic, a median of the squared pairwise distances. In each system, a corresponding distance metric is used to ensure that global symmetries aren't picked up by the diffusion process. To do this, we find the group operations on the system that leave the Hamiltonian invariant. This will then leave the system's corresponding system dynamics unchanged.

For the Ising model there exists a $\mathbb{Z}_2$ symmetry action $s \to -s$. For lattice configurations s^A, s^B we may use the invariant distance metric

$$d(s^A, s^B) = \min(|s^A - s^B|, |s^A + s^B|) \tag{7}$$

For the 2D XY model there exists global $O(2)$ symmetry actions $\theta \to \pm\theta + \Delta\Theta$. For lattice configurations θ^A, θ^B we may use the invariant distance metric

$$d(\theta_A, \theta_B) = N - \max\left(\left|\sum_j e^{i(\theta_j^A - \theta_j^B)}\right|, \left|\sum_j e^{i(\theta_j^A + \theta_j^B)}\right|\right) \tag{8}$$

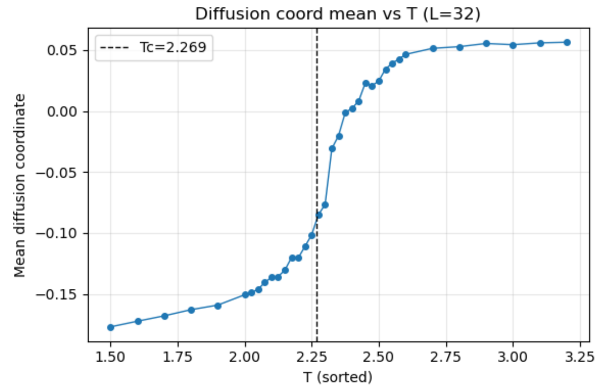


FIG. 1. The first diffusion coordinate increases monotonically with T. Notice that the region of the highest slope diagrammatically corresponds to the presence of the critical temperature.

VI. RESULTS

A. Ising Model Results

a. Cluster Analysis The silhouette score provides a label-free measure of clustering quality: it compares typical within-cluster distances to nearest-cluster distances, with larger values indicating better-separated and more cohesive clusters [16]. For the selected kernel scale, silhouette scores evaluated over $k \in \{2, \ldots, 8\}$ are maximized at $k = 2$, consistent with the expected two-phase structure of the Ising model. Additionally we see that the largest gap occurs between $\lambda_1 - \lambda_2$. This suggests that the number of emerging order parameters is also a function of the number of eigenvalues taken.

b. Embedding Space Figure 2 shows the central results. Configurations form a curved cloud surface in diffusion coordinates. Low temperatures sit on one side and high temperatures on the other. Clustering in this space produces a clean ordered/disordered split. Plotting the fraction of points assigned to the ordered cluster vs. T yields a rapid drop near the known T_c [14]. In Figure 4 see that the estimated T_c is found to be around $T_c \approx 2.3$, which matches well with the expected value.

Two-dimensional projections can obscure separations that are clearer in higher-dimensional embeddings. Including additional diffusion coordinates (notably Ψ_2 and Ψ_3) yields a visibly tighter separation between the two clusters when visualized in three dimensions (Figure 3), indicating that the subleading diffusion modes carry additional geometric information beyond the leading coordinate.

c. Learned Order Parameter The first nontrivial diffusion coordinate $\Psi_1 = \lambda_1\psi_1$ typically captures the slowest mode of the random walk [4]. In the Ising data, this mode closely tracks the ordered–disordered change, so Ψ_1 varies monotonically with T (Figure 1).

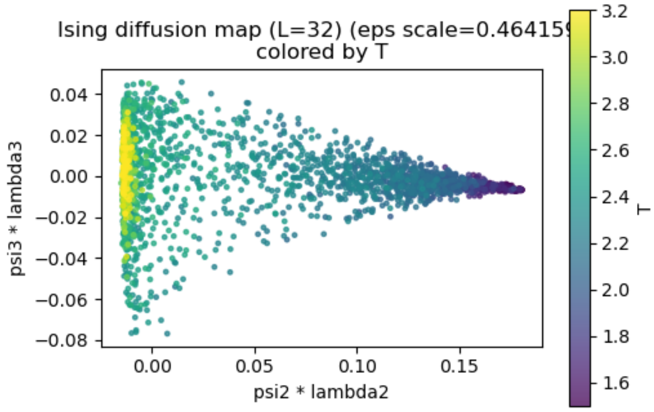


FIG. 2. Two-dimensional diffusion embedding colored by temperature T for the Ising Model at $L = 32$.

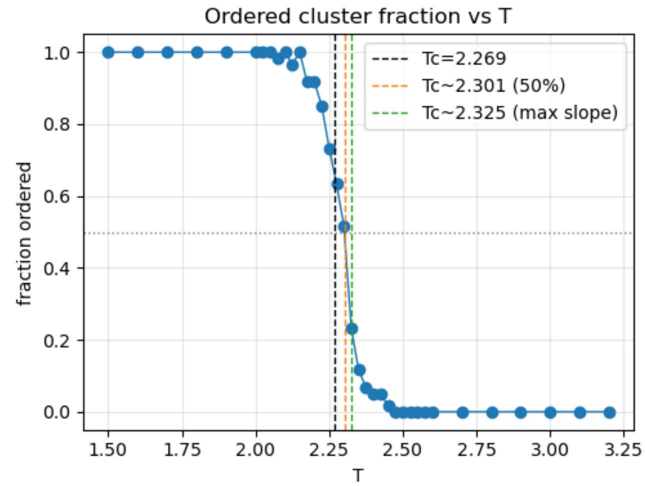


FIG. 4. Cluster Fraction as a function of temperature $L = 32$

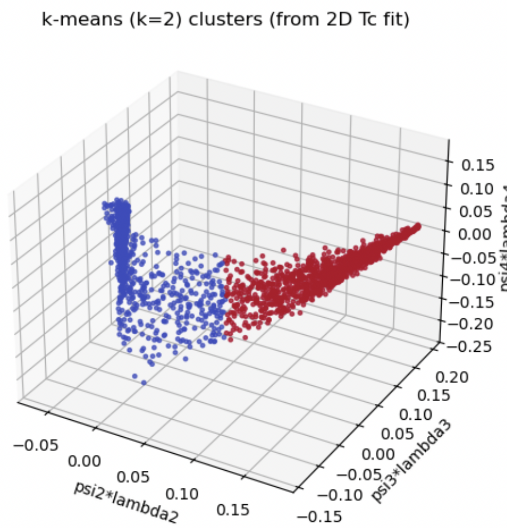


FIG. 3. Three-dimensional diffusion embedding using (Ψ_2, Ψ_3, Ψ_4) at $L = 32$.

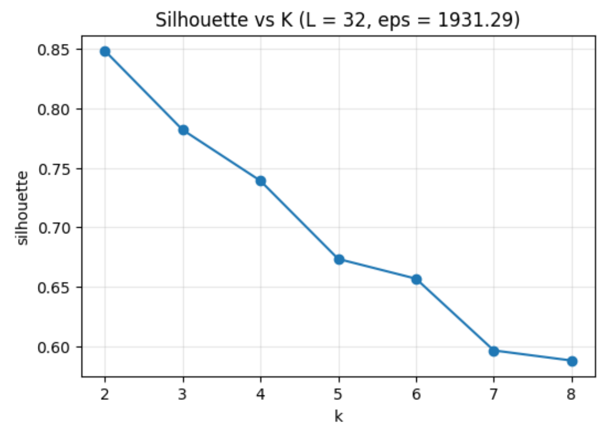


FIG. 5. XY Model Silhouette Score vs K Cluster Number ($L = 32$).

d. Finite-Size + Critical Temperature Check: Finally, the unsupervised pipeline is applied separately for each lattice size ($L = 64, 100, 200$) and the ordered-fraction curves are overlaid. The curves sharpen with increasing L, and the extracted pseudo-critical temperature $T_c(L)$ drifts toward the exact T_c as $1/L$ decreases, which is the expected trend [1].

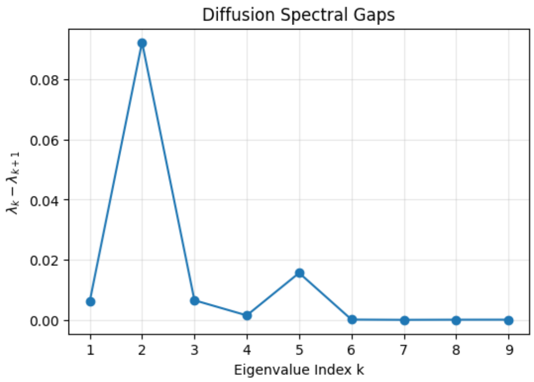
B. 2D XY Model Results

For the selected kernel scale, silhouette scores are similarly maximized at $k = 2$ (Figure 1), consistent with the expected two-phase structure of the XY model. Additionally, the spectral gaps are plotted for both embeddings in Figure 6, showing distinct behavior for each embedding. In particular, we see that the rotational symmetry picks up an additional eigenvalue useful for classification for the Euclidean embedding, which is then promptly discarded for the $O(2)$ invariant embedding.

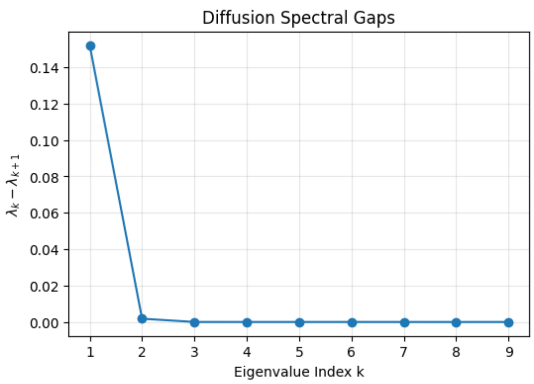
a. Separation of Phases Figure 7 shows the central results. Using the Euclidean distance, clustering is much harder because a nonlinear relationship emerges between temperature and the diffusion coordinates. This stems from the model's ability to capture rotational symmetry. We can use the $O(2)$ invariant kernel to get a clearer order parameter. Clustering in this space produces a clean, ordered/disordered split, indicating proper capture of the topological phase transition.

Additionally, a 3D representation of the embedding space is provided, showing low variation along the third diffusion coordinate, indicating that most of the structure arises in the first two coordinates.

b. Discovered Order Parameter Although there are no local order parameters in the XY model, we see that

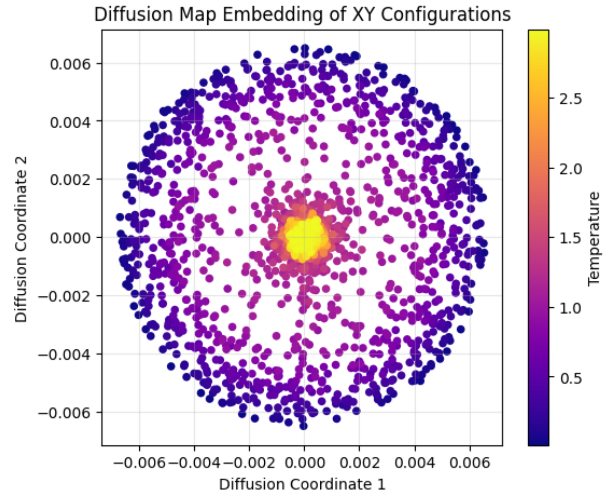


(a) Euclidean Embedding XY 32×32 Lattice

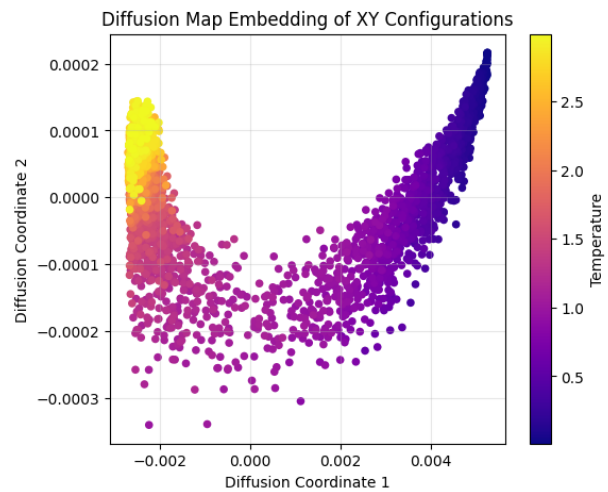


(b) O(2) Invariant Embedding XY 32×32 Lattice

FIG. 6. Eigengap Measurements $\lambda_k - \lambda_{k+1}$ for both the Euclidean and O(2)-invariant embeddings



(a) Euclidean Embedding XY 32×32 Lattice



(b) O(2) Invariant Embedding XY 32×32 Lattice

FIG. 7. Comparison of Diffusion-Map embeddings for lattice configurations between Euclidean and O(2)-Invariant distance metrics for 2D XY Model on a 32×32 lattice.

Ψ_1 varies monotonically with T. Moreover that abrupt changes in Ψ_1 correspond to sudden changes in the structure of the lattice indicative of a phase transition. (Figure 9).

c. Finite-size check. Finally, the unsupervised pipeline is applied separately for each lattice size ($L = 32, 64, 128$), as shown in Figure 10. Despite incorrectly obtaining the critical temperature, the points follow an inverse log-like relationship, and the extracted pseudo-critical temperature $T_c(L)$ drifts toward the exact T_c as $1/\log^2(L)$ decreases, which is the expected trend [1]. This further emphasizes the importance of finite-size checks, with strong effects that prevent the correct location of the critical temperature.

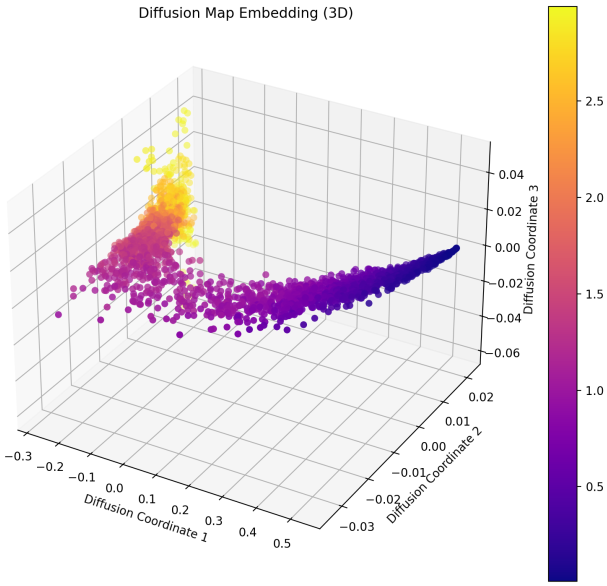
VII. LIMITATIONS & FUTURE WORK

Compared to many "black box" approaches, diffusion maps give two practical advantages:

- **Interpretability:** the embedding is visible, and the random-walk picture explains why two phases appear as two weakly connected regions [4, 9].
- **A direct transition curve:** clustering in the embedding gives scalar diagnostica (ordered fraction, order parameter gradient) whose sharp change marks the transition, without training a neural network.

At the same time, the approach is not parameter-free. The analysis still requires choosing a representation (raw spins vs block-averaged spins), a graph construction (k), and a kernel scale ε. Those choices can change the overall picture and diagnostics received. The purpose of this pipeline is therefore not to claim an optimal method, but to provide a controlled benchmark for explicitly examining sensitivity to these choices.

Additionally, we have not explored the use of different symmetry-invariant distance metrics. Indeed, the choice of a distance metric matters greatly in representing latent embeddings. Prior work [15] has produced a diffusion map that converges to the Laplace-Beltrami operator, which is well-defined and properly behaved; future work



(a) colored by temperature

FIG. 8. Three-dimensional diffusion embedding using (Ψ_2, Ψ_3, Ψ_4) at $L = 32$.

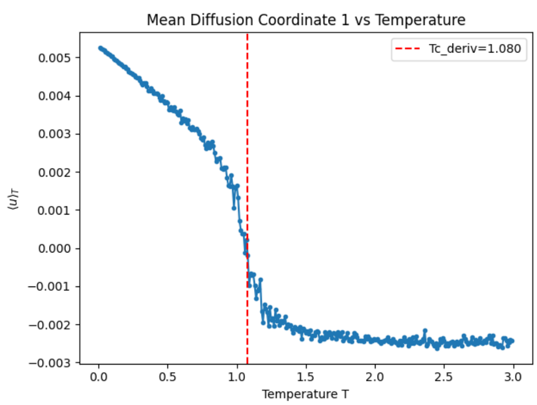


FIG. 9. The mean first diffusion coordinate Ψ_1 is plotted as a function of temperature. To obtain the estimated critical temprerature we find when the derivative has maximal norm, which occurs at $T_{\text{est}} \approx 1.08$

can explore combining this with the analysis of lattice configurations. Looking ahead, the same unsupervised pipeline can be applied to models with much more exotic transitions or that contain multiple order parameters and phase transitions.

VIII. CONCLUSION

A reproducible pipeline is presented for unsupervised phase-transition detection on both the 2D Ising model and 2D XY model using diffusion maps and simple clustering. We present modification of diffusion

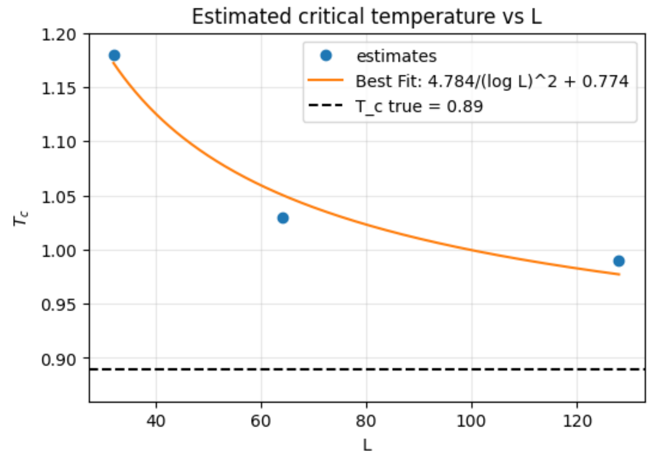


FIG. 10. The estimated critical temperature is plotted as a function of the lattice sizes L used. We see that the predicted points follow a $1/\log(L)^2$ curve

kernels which are invariant under group actions of the corresponding Hamiltonian. These kernels more easily capture the underlying structure and order parameter which remain monotone with respect to the original model parameters. The principal outcome is an interpretable embedding and a scalar ordered-fraction diagnostic whose rapid change identifies the transition region, together with a basic finite-size sanity check. The resulting methodology provides a controlled baseline against which more subtle transitions (e.g., BKT physics) can be evaluated in subsequent stages. Future work in this direction could include analyzing more complicated systems such as the Generalized 2D XY Model [7]. Future work also permits extending these systems to more realistic quantum systems that exhibit topological quantum phases.

IX. REPRODUCIBILITY

- The implementations used to generate data and figures are provided as supplementary material.
- The workflow is specified by the algorithmic descriptions in Sections V A–V C

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